



Qualification Pack



Surveyor

QP Code: CON/Q0902

Version: 4.0

NSQF Level: 5.5

Construction Skill Development Council of India || CPB 103 & 104 (1st Floor), Block 4B, DLF Corporate Park, Phase III, MG Road
Gurgaon-122002 || email: abhisek@csdcindia.org



Qualification Pack

Contents

CON/Q0902: Surveyor	3
<i>Brief Job Description</i>	3
Applicable National Occupational Standards (NOS)	3
<i>Compulsory NOS</i>	3
<i>Qualification Pack (QP) Parameters</i>	3
CON/N0903: Carry out temporary adjustment of survey instruments as per standard methods	5
CON/N0904: Conduct linear measurements using survey instruments and tools	9
CON/N0905: Carry out leveling and cross sectioning survey	15
CON/N0906: Carry out setting out operations for buildings/ other structures	21
CON/N0907: Carry out topographic survey	26
CON/N9002: Manage workplace for safe and healthy work environment	31
CON/N8003: Supervise, monitor and evaluate performance of subordinates at workplace	37
DGT/VSQ/N0103: Employability Skills (90 Hours)	41
Assessment Guidelines and Weightage	49
<i>Assessment Guidelines</i>	49
<i>Assessment Weightage</i>	50
Acronyms	52
Glossary	55



Qualification Pack

CON/Q0902: Surveyor

Brief Job Description

A Surveyor is responsible for carrying out land surveying using modern and conventional systems and instruments at construction sites.

Personal Attributes

The individual should be physically fit to work in different locations and conditions. The person should be organized, diligent, methodical, safety-conscious, and a prompt decision-maker. The individual should have good communication skills.

Applicable National Occupational Standards (NOS)

Compulsory NOS:

1. [CON/N0903: Carry out temporary adjustment of survey instruments as per standard methods](#)
2. [CON/N0904: Conduct linear measurements using survey instruments and tools](#)
3. [CON/N0905: Carry out leveling and cross sectioning survey](#)
4. [CON/N0906: Carry out setting out operations for buildings/ other structures](#)
5. [CON/N0907: Carry out topographic survey](#)
6. [CON/N9002: Manage workplace for safe and healthy work environment](#)
7. [CON/N8003: Supervise, monitor and evaluate performance of subordinates at workplace](#)
8. [DGT/VSQ/N0103: Employability Skills \(90 Hours\)](#)

Qualification Pack (QP) Parameters

Sector	Construction
Sub-Sector	Real Estate and Infrastructure construction
Occupation	Surveying
Country	India
NSQF Level	5.5



Qualification Pack

Credits	20
Aligned to NCO/ISCO/ISIC Code	NCO-2015/2165.0200
Minimum Educational Qualification & Experience	UG in relevant field (Completed 3rd year of 3-year / 4-year UG in Civil Engineering) OR Completed 2nd year diploma after 12th (in Civil Engineering) with 1 Year of experience in relevant industry OR Completed 2nd year of UG (UG Diploma) (in Civil Engineering) with 1 Year of experience in relevant industry OR Completed 3 year diploma after 10th (in Civil Engineering) with 2 Years of experience in relevant industry OR 12th grade Pass with 3 Years of experience in relevant industry
Minimum Level of Education for Training in School	12th Class
Pre-Requisite License or Training	NIL
Minimum Job Entry Age	18 Years
Last Reviewed On	NA
Next Review Date	30/04/2028
NSQC Approval Date	08/05/2025
Version	4.0
Reference code on NQR	QG-5.5-CO-03966-2025-V2-CSDCI
NQR Version	4.0



Qualification Pack

CON/N0903: Carry out temporary adjustment of survey instruments as per standard methods

Description

This unit describes the skills and knowledge required to carry out temporary adjustments of the survey instruments as per standard methods.

Scope

The scope covers the following :

- Carry out initial set up, centering and levelling of survey equipment (levels, theodolite, total station etc.)

Elements and Performance Criteria

Carry out initial set up, centering and levelling of survey equipment (levels, theodolite, total station etc.)

To be competent, the user/individual on the job must be able to:

- PC1.** locate the station mark on the ground surface
- PC2.** place the tripod over the station mark
- PC3.** fix the instrument on the tripod head
- PC4.** level the instrument by adjusting the legs of tripod
- PC5.** ensure that the legs of tripod are sufficient distance apart and are pressed firmly on the ground to provide stability to the instrument
- PC6.** ensure that the height of instrument is such that it is convenient to take readings
- PC7.** perform the centering of the instrument over the station mark by using plumb bob, optical plummet or laser pointer as per applicability
- PC8.** adjust the levels of horizontal and vertical axis of the instruments using standard references/indicators such as bubble tubes, spirit level and adjustable screws as per standard practices
- PC9.** focus the telescope on the concerned object or point for clear visibility, by adjusting the eyepiece/ knobs/screw and cross-hair

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

- KU1.** standard practices for surveying works
- KU2.** safety rules and regulations for handling and storing relevant tools, equipment, and materials required for relevant works in accordance with organizational norms
- KU3.** importance of personal protection including the use of related safety gears and equipment in accordance with organizational norms
- KU4.** service request procedures for tools, materials and equipment as per organizational norms
- KU5.** procedure for maintenance of tools and equipment



Qualification Pack

- KU6.** statutory compliance requirement related to workmen engagement
- KU7.** different survey instruments and their temporary adjustments
- KU8.** basic concepts of temporary and permanent adjustments, its purpose and importance
- KU9.** procedures for carrying out temporary adjustments
- KU10.** procedure for centering and levelling in survey instruments
- KU11.** various types of permanent adjustments
- KU12.** types of errors in surveying instruments

Generic Skills (GS)

User/individual on the job needs to know how to:

- GS1.** write in at least one language, preferably in the local language of the site
- GS2.** read sketches, instructions provided for the work, and various signboards, safety rules, safety tags, exit route information in one or more languages, preferably in the local language of the site
- GS3.** speak in one or more language, preferably one of the local language at the site
- GS4.** communicate orally and effectively with team members
- GS5.** analyze the safety aspect of the workplace
- GS6.** plan work and organize required resource effectively
- GS7.** complete work as per agreed time schedule and quality parameters
- GS8.** resolve any conflict within the teammates
- GS9.** evaluate the complexity of the tasks
- GS10.** identify any violation of safety norms during the work



Qualification Pack

Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Carry out initial set up, centering and levelling of survey equipment (levels, theodolite, total station etc.)</i>	30	70	-	-
PC1. locate the station mark on the ground surface	-	-	-	-
PC2. place the tripod over the station mark	-	-	-	-
PC3. fix the instrument on the tripod head	-	-	-	-
PC4. level the instrument by adjusting the legs of tripod	-	-	-	-
PC5. ensure that the legs of tripod are sufficient distance apart and are pressed firmly on the ground to provide stability to the instrument	-	-	-	-
PC6. ensure that the height of instrument is such that it is convenient to take readings	-	-	-	-
PC7. perform the centering of the instrument over the station mark by using plumb bob, optical plummet or laser pointer as per applicability	-	-	-	-
PC8. adjust the levels of horizontal and vertical axis of the instruments using standard references/indicators such as bubble tubes, spirit level and adjustable screws as per standard practices	-	-	-	-
PC9. focus the telescope on the concerned object or point for clear visibility, by adjusting the eyepiece/ knobs/screw and cross-hair	-	-	-	-
NOS Total	30	70	-	-



Qualification Pack

National Occupational Standards (NOS) Parameters

NOS Code	CON/N0903
NOS Name	Carry out temporary adjustment of survey instruments as per standard methods
Sector	Construction
Sub-Sector	Real Estate and Infrastructure construction
Occupation	Surveying
NSQF Level	5.5
Credits	3
Version	4.0
Last Reviewed Date	08/05/2025
Next Review Date	30/04/2028
NSQC Clearance Date	08/05/2025



Qualification Pack

CON/N0904: Conduct linear measurements using survey instruments and tools

Description

This unit describes the skills and knowledge required to conduct linear measurements using survey instruments and tools

Scope

The scope covers the following :

- Conduct linear measurements using chain and measuring tape for measurement in plans and slopes
- Conduct linear measurement using total stations

Elements and Performance Criteria

Conduct linear measurements using chain and measuring tape for measurement in plans and slopes

To be competent, the user/individual on the job must be able to:

- PC1.** interpret site drawings, layout plans, and boundary maps etc
- PC2.** estimate the optimal length of chain required for measuring the required distance
- PC3.** select suitable tools, instruments and marking materials for conducting required work
- PC4.** identify the start point/ benchmark for measurement as per work requirement
- PC5.** monitor subordinates to ensure that the unfolding of the chain or tape is as per standard practice
- PC6.** check that the subordinates have collected appropriate tools such as arrows, pegs etc. in required number
- PC7.** check that the ranging rods are vertical and of appropriate length to avoid any errors in ranging
- PC8.** ensure that the chain or tape is stretched appropriately to avoid any errors due to sagging of the same
- PC9.** coordinate between leader and follower to maintain straight line of measurement and proper placement of flags or pegs
- PC10.** ensure that the peg is fixed at all station points of the survey
- PC11.** record the chainages or measured distance for respective points using proper symbols in the field book, as per standard practice
- PC12.** instruct subordinates to use appropriate tools for transferring points on ground/ slopes

Conduct linear measurement using total stations

To be competent, the user/individual on the job must be able to:

- PC13.** identify the location of master and secondary control points to set up the instrument
- PC14.** identify the suitable benchmark as a reference to obtain a back-sight at station point
- PC15.** locate the exact position of the instrument with respect to the back-sight and foresight points
- PC16.** set up the instrument exactly over the station point
- PC17.** carry out temporary adjustments of the instrument



Qualification Pack

- PC18.** input various data/parameters as per job details, locations and technical specifications
- PC19.** identify the location of staff point for measuring the horizontal distance as per the drawings or co-ordinates
- PC20.** instruct the subordinate to hold the staff exactly at the staff point
- PC21.** operate the total station instrument to compute and record the required data
- PC22.** remove the data card from the instrument and transfer the same into compatible computer

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

- KU1.** standard practices for surveying works
- KU2.** safety rules and regulations for handling and storing relevant tools, equipment, and materials required for relevant works in accordance with organizational norms
- KU3.** importance of personal protection including the use of related safety gears and equipment in accordance with organizational norms
- KU4.** service request procedures for tools, materials and equipment as per organizational norms
- KU5.** procedure for maintenance of tools and equipment
- KU6.** different types of chains and tapes based upon their length and material
- KU7.** how to read and understand site drawings, layout plans, boundary maps etc
- KU8.** procedure for calculations of dimensions and angles
- KU9.** concept of latitudes and longitudes, their computations and use
- KU10.** use of various tools and materials in surveying and their application
- KU11.** how to check the serviceability of various tools and tackles
- KU12.** different causes of errors in the linier measurements, their impact on project measurement and how to avoid such errors
- KU13.** various difficulties that may be faced during survey and procedure to overcome the same
- KU14.** procedure for conducting linier measurements on site
- KU15.** standard hand signals, their interpretations and applications
- KU16.** how to maintain and store the various tools and tackles
- KU17.** how to document readings in field book for different types of surveys
- KU18.** ideal conditions for selecting a station point
- KU19.** concept of bench mark, types, its importance in surveying, ideal conditions for selecting a bench mark
- KU20.** survey concrete pillars
- KU21.** principal of operation of total station
- KU22.** various applications of total station
- KU23.** how to feed and retrieve data from the total station
- KU24.** how to transfer data from instrument to computer
- KU25.** how to operate computer
- KU26.** standard procedure for operating total stations
- KU27.** type of reflective staff/prism used in survey



Qualification Pack

- KU28.** standard procedure for conducting liner measurements with total station
- KU29.** how to download Field Data through various sources like; Data Cable, Data Card, CD, Pen Drive etc.
- KU30.** procedure of conducting as built measurements
- KU31.** recording and plotting of as built measurements

Generic Skills (GS)

User/individual on the job needs to know how to:

- GS1.** write in at least one language, preferably in the local language of the site
- GS2.** read sketches, instructions provided for the work, and various signboards, safety rules, safety tags, exit route information in one or more languages, preferably in the local language of the site
- GS3.** speak in one or more language, preferably one of the local language at the site
- GS4.** communicate orally and effectively with team members
- GS5.** analyze the safety aspect of the workplace
- GS6.** plan work and organize required resource effectively
- GS7.** complete work as per agreed time schedule and quality parameters
- GS8.** resolve any conflict within the teammates
- GS9.** evaluate the complexity of the tasks
- GS10.** identify any violation of safety norms during the work



Qualification Pack

Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Conduct linear measurements using chain and measuring tape for measurement in plans and slopes</i>	15	35	-	-
PC1. interpret site drawings, layout plans, and boundary maps etc	-	-	-	-
PC2. estimate the optimal length of chain required for measuring the required distance	-	-	-	-
PC3. select suitable tools, instruments and marking materials for conducting required work	-	-	-	-
PC4. identify the start point/ benchmark for measurement as per work requirement	-	-	-	-
PC5. monitor subordinates to ensure that the unfolding of the chain or tape is as per standard practice	-	-	-	-
PC6. check that the subordinates have collected appropriate tools such as arrows, pegs etc. in required number	-	-	-	-
PC7. check that the ranging rods are vertical and of appropriate length to avoid any errors in ranging	-	-	-	-
PC8. ensure that the chain or tape is stretched appropriately to avoid any errors due to sagging of the same	-	-	-	-
PC9. coordinate between leader and follower to maintain straight line of measurement and proper placement of flags or pegs	-	-	-	-
PC10. ensure that the peg is fixed at all station points of the survey	-	-	-	-
PC11. record the chainages or measured distance for respective points using proper symbols in the field book, as per standard practice	-	-	-	-
PC12. instruct subordinates to use appropriate tools for transferring points on ground/ slopes	-	-	-	-
<i>Conduct linear measurement using total stations</i>	15	35	-	-



Qualification Pack

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC13. identify the location of master and secondary control points to set up the instrument	-	-	-	-
PC14. identify the suitable benchmark as a reference to obtain a back-sight at station point	-	-	-	-
PC15. locate the exact position of the instrument with respect to the back-sight and foresight points	-	-	-	-
PC16. set up the instrument exactly over the station point	-	-	-	-
PC17. carry out temporary adjustments of the instrument	-	-	-	-
PC18. input various data/parameters as per job details, locations and technical specifications	-	-	-	-
PC19. identify the location of staff point for measuring the horizontal distance as per the drawings or co-ordinates	-	-	-	-
PC20. instruct the subordinate to hold the staff exactly at the staff point	-	-	-	-
PC21. operate the total station instrument to compute and record the required data	-	-	-	-
PC22. remove the data card from the instrument and transfer the same into compatible computer	-	-	-	-
NOS Total	30	70	-	-



Qualification Pack

National Occupational Standards (NOS) Parameters

NOS Code	CON/N0904
NOS Name	Conduct linear measurements using survey instruments and tools
Sector	Construction
Sub-Sector	Real Estate and Infrastructure construction
Occupation	Surveying
NSQF Level	5.5
Credits	3
Version	4.0
Last Reviewed Date	08/05/2025
Next Review Date	30/04/2028
NSQC Clearance Date	08/05/2025



Qualification Pack

CON/N0905: Carry out leveling and cross sectioning survey

Description

This unit describes the skills and knowledge required to carry out leveling and cross sectioning survey

Scope

The scope covers the following :

- Carry out leveling and cross section survey using total station

Elements and Performance Criteria

Carry out leveling and cross section survey using total station

To be competent, the user/individual on the job must be able to:

- PC1.** interpret the scope of survey and data to be collected
- PC2.** identify and locate the first station point to set up the instrument
- PC3.** identify and locate the benchmark on the field as instructed by the surveyor
- PC4.** setup the instrument exactly over the station point
- PC5.** carry out temporary adjustments of the instrument
- PC6.** input data regarding temperature and pressure bars into the total station
- PC7.** input the data regarding survey job, station point, type of measurement, RL of benchmark etc. as required by the instrument
- PC8.** instruct the subordinate to hold the reflective staff exactly over the benchmark of known elevation
- PC9.** bisect the reflective prism mounted on the staff with the help of tangent screw and fine tune screw
- PC10.** bisect the prism using the input panel on the total station to compute the reduced level at the station point from the RL earlier fed in the instrument using the telescope
- PC11.** identify the direction and position of the terminal point/ station and place the staff
- PC12.** obtain the RL(reduced level) of the staff point by bisecting the reflective prism and instructing the instrument to calculate the RL
- PC13.** follow the standard procedure to obtain readings at multiple staff points at fixed interval
- PC14.** transfer the instrument to the next point as per requirement and take a back sight reading at the previous intermediate point or staff point
- PC15.** take staff readings at fixed interval on multiple points normal to the center line on both sides
- PC16.** check the display of the instrument to confirm that data displayed is as per requirement
- PC17.** remove the data card from the instrument and transfer the same into compatible computer

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:



Qualification Pack

- KU1.** standard practices for surveying works
- KU2.** safety rules and regulations for handling and storing relevant tools, equipment, and materials required for relevant works in accordance with organizational norms
- KU3.** importance of personal protection including the use of related safety gears and equipment in accordance with organizational norms
- KU4.** service request procedures for tools, materials and equipment as per organizational norms
- KU5.** procedure for maintenance of tools and equipment
- KU6.** statutory compliance requirement related to workmen engagement
- KU7.** how to read and understand the layout plan, alignment diagram, and other drawings
- KU8.** concept and principles of leveling, different types of leveling, their application
- KU9.** standard procedure for conducting leveling works
- KU10.** importance of leveling in various sub sectors
- KU11.** different instruments used for leveling
- KU12.** different parts and components of the total station
- KU13.** interpretation of site drawings, layout plans, boundary maps etc.
- KU14.** calculations for computations of dimensions and angles
- KU15.** concept of latitudes and longitudes, their computations and use
- KU16.** various tools, materials and their use and applications in surveying
- KU17.** how to check the serviceability of various tools and tackles
- KU18.** different causes of errors in the leveling works, their impact on the project
- KU19.** different terminologies used in the leveling operations
- KU20.** procedures for making entries in the field book and making necessary calculations
- KU21.** ideal conditions for selecting a bench mark
- KU22.** procedure for laying slopes and gradients for roads, bridges, pipelines, canals etc
- KU23.** errors in slope alignment and their implications
- KU24.** calculation of RL using rise and fall method and height of collimation method, and performing necessary checks
- KU25.** how to feed and retrieve data from the total station
- KU26.** how to transfer data from instrument to computer
- KU27.** computer literacy
- KU28.** standard procedure for operating total stations
- KU29.** type of reflective staff used in survey
- KU30.** placing of measurement/ reflective staff at appropriate location

Generic Skills (GS)

User/individual on the job needs to know how to:

- GS1.** write in at least one language, preferably in the local language of the site
- GS2.** read sketches, instructions provided for the work, and various signboards, safety rules, safety tags, exit route information in one or more languages, preferably in the local language of the site



Qualification Pack

- GS3.** speak in one or more language, preferably one of the local language at the site
- GS4.** communicate orally and effectively with team members
- GS5.** analyze the safety aspect of the workplace
- GS6.** plan work and organize required resource effectively
- GS7.** complete work as per agreed time schedule and quality parameters
- GS8.** resolve any conflict within the teammates
- GS9.** evaluate the complexity of the tasks
- GS10.** identify any violation of safety norms during the work



Qualification Pack

Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Carry out leveling and cross section survey using total station</i>	30	70	-	-
PC1. interpret the scope of survey and data to be collected	-	-	-	-
PC2. identify and locate the first station point to set up the instrument	-	-	-	-
PC3. identify and locate the benchmark on the field as instructed by the surveyor	-	-	-	-
PC4. setup the instrument exactly over the station point	-	-	-	-
PC5. carry out temporary adjustments of the instrument	-	-	-	-
PC6. input data regarding temperature and pressure bars into the total station	-	-	-	-
PC7. input the data regarding survey job, station point, type of measurement, RL of benchmark etc. as required by the instrument	-	-	-	-
PC8. instruct the subordinate to hold the reflective staff exactly over the benchmark of known elevation	-	-	-	-
PC9. bisect the reflective prism mounted on the staff with the help of tangent screw and fine tune screw	-	-	-	-
PC10. bisect the prism using the input panel on the total station to compute the reduced level at the station point from the RL earlier fed in the instrument using the telescope	-	-	-	-
PC11. identify the direction and position of the terminal point/ station and place the staff	-	-	-	-
PC12. obtain the RL(reduced level) of the staff point by bisecting the reflective prism and instructing the instrument to calculate the RL	-	-	-	-



Qualification Pack

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC13. follow the standard procedure to obtain readings at multiple staff points at fixed interval	-	-	-	-
PC14. transfer the instrument to the next point as per requirement and take a back sight reading at the previous intermediate point or staff point	-	-	-	-
PC15. take staff readings at fixed interval on multiple points normal to the center line on both sides	-	-	-	-
PC16. check the display of the instrument to confirm that data displayed is as per requirement	-	-	-	-
PC17. remove the data card from the instrument and transfer the same into compatible computer	-	-	-	-
NOS Total	30	70	-	-



Qualification Pack

National Occupational Standards (NOS) Parameters

NOS Code	CON/N0905
NOS Name	Carry out leveling and cross sectioning survey
Sector	Construction
Sub-Sector	Real Estate and Infrastructure construction
Occupation	Surveying
NSQF Level	5.5
Credits	3
Version	4.0
Last Reviewed Date	08/05/2025
Next Review Date	30/04/2028
NSQC Clearance Date	08/05/2025



Qualification Pack

CON/N0906: Carry out setting out operations for buildings/ other structures

Description

This unit describes the skills and knowledge required to carry out setting out for buildings/ other structures

Scope

The scope covers the following :

- Perform preparatory activities prior to setting out
- Carry out setting out operations

Elements and Performance Criteria

Perform preparatory activities prior to setting out

To be competent, the user/individual on the job must be able to:

- PC1.** obtain work instructions, plans, specifications, quality requirements and operational details
- PC2.** plan the sequence to be adopted for setting out works
- PC3.** check the serviceability of the required tools and instruments and report any errors/ faults to the seniors
- PC4.** estimate the required materials and ensure that they comply with work requirements

Carry out setting out operations

To be competent, the user/individual on the job must be able to:

- PC5.** locate and identify the survey pegs or control point and mark the boundary lines
- PC6.** calculate the distance and direction of building line from the boundary or base line as per plans/drawings
- PC7.** carry out liner and angular measurements to locate the building line on field from the boundary or baseline
- PC8.** identify all the grid lines /numbers provided on the plans/drawings and establish them on the ground using string (line dori) and pegs
- PC9.** conduct the erection/construction of profile/hurdles/platform at the both ends of grid lines and mark them as per their grid numbers
- PC10.** determine the corner of building on set building line to true measurement from adjacent boundary and mark the same as per drawings and specifications
- PC11.** set out right angle to the building line from the identified corner point using TS, theodolite or triangulation principals
- PC12.** recheck all the measurements accurately using TS or triangulation methods and confirm the same with drawings

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:



Qualification Pack

- KU1.** standard practices for surveying works
- KU2.** safety rules and regulations for handling and storing relevant tools, equipment, and materials required for relevant works in accordance with organizational norms
- KU3.** importance of personal protection including the use of related safety gears and equipment in accordance with organizational norms
- KU4.** service request procedures for tools, materials and equipment as per organizational norms
- KU5.** procedure for maintenance of tools and equipment
- KU6.** statutory compliance requirement related to workmen engagement
- KU7.** application and requirements for line, level and plumb in construction projects
- KU8.** basic mathematical techniques associated with setting out
- KU9.** setting out techniques
- KU10.** method and sequence of setting out of the building profile on the ground
- KU11.** site isolation and traffic control responsibilities and authorities
- KU12.** types, characteristics, technical capabilities and limitations of setting out devices
- KU13.** how to read and understand the layout plan, alignment diagram, and other drawings
- KU14.** concept and principles of leveling, different types of leveling, their application
- KU15.** standard procedure for conducting leveling works
- KU16.** importance of leveling in various sub sectors
- KU17.** different instruments used for leveling
- KU18.** different parts and components of the instruments
- KU19.** interpreting site drawings, layout plans, boundary maps etc.
- KU20.** calculations for computations of dimensions and angle

Generic Skills (GS)

User/individual on the job needs to know how to:

- GS1.** write in at least one language, preferably in the local language of the site
- GS2.** read sketches, instructions provided for the work, and various signboards, safety rules, safety tags, exit route information in one or more languages, preferably in the local language of the site
- GS3.** speak in one or more language, preferably one of the local language at the site
- GS4.** communicate orally and effectively with team members
- GS5.** analyze the safety aspect of the workplace
- GS6.** plan work and organize required resource effectively
- GS7.** complete work as per agreed time schedule and quality parameters
- GS8.** resolve any conflict within the teammates
- GS9.** evaluate the complexity of the tasks
- GS10.** identify any violation of safety norms during the work



Qualification Pack

Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Perform preparatory activities prior to setting out</i>	15	30	-	-
PC1. obtain work instructions, plans, specifications, quality requirements and operational details	-	-	-	-
PC2. plan the sequence to be adopted for setting out works	-	-	-	-
PC3. check the serviceability of the required tools and instruments and report any errors/ faults to the seniors	-	-	-	-
PC4. estimate the required materials and ensure that they comply with work requirements	-	-	-	-
<i>Carry out setting out operations</i>	15	40	-	-
PC5. locate and identify the survey pegs or control point and mark the boundary lines	-	-	-	-
PC6. calculate the distance and direction of building line from the boundary or base line as per plans/drawings	-	-	-	-
PC7. carry out liner and angular measurements to locate the building line on field from the boundary or baseline	-	-	-	-
PC8. identify all the grid lines /numbers provided on the plans/drawings and establish them on the ground using string (line dori) and pegs	-	-	-	-
PC9. conduct the erection/construction of profile/hurdles/platform at the both ends of grid lines and mark them as per their grid numbers	-	-	-	-
PC10. determine the corner of building on set building line to true measurement from adjacent boundary and mark the same as per drawings and specifications	-	-	-	-
PC11. set out right angle to the building line from the identified corner point using TS, theodolite or triangulation principals	-	-	-	-



Qualification Pack

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC12. recheck all the measurements accurately using TS or triangulation methods and confirm the same with drawings	-	-	-	-
NOS Total	30	70	-	-



Qualification Pack

National Occupational Standards (NOS) Parameters

NOS Code	CON/N0906
NOS Name	Carry out setting out operations for buildings/ other structures
Sector	Construction
Sub-Sector	Real Estate and Infrastructure construction
Occupation	Surveying
NSQF Level	5.5
Credits	3
Version	4.0
Last Reviewed Date	08/05/2025
Next Review Date	30/04/2028
NSQC Clearance Date	08/05/2025



Qualification Pack

CON/N0907: Carry out topographic survey

Description

This unit describes the skills and knowledge required to carry out topographic survey

Scope

The scope covers the following :

- Carry out topographic survey
- Prepare contour and topographic map

Elements and Performance Criteria

Carry out topographic survey

To be competent, the user/individual on the job must be able to:

- PC1.** identify the boundaries of the area to be surveyed
- PC2.** establish control points/bench marks to initiate the survey
- PC3.** identify a suitable location for setting up a total station such that maximum points or features are visible from this point
- PC4.** set up the instrument at identified point and perform necessary adjustments
- PC5.** take the back sight from instrument point to BM and set the readings as zero for this point
- PC6.** instruct the subordinate to place the staff at all relevant feature points like corners of buildings, trees, pillars etc
- PC7.** operate the survey instrument to focus on the staff and record the required readings for the fore sights/intermediate sight as per the requirement
- PC8.** coordinate with team members to change the instrument placing and repeat the same procedure for taking further reading
- PC9.** transfer/upload the recorded data on the computer system using appropriate CAD software

Prepare contour and topographic map

To be competent, the user/individual on the job must be able to:

- PC10.** obtain approval from seniors regarding correctness of the collected data
- PC11.** segregate the required information in desired formats using available software
- PC12.** decide the scale for plotting the survey data
- PC13.** plot the recorded points using the CAD software to get the required contour/ topographic map as per the measured distance, location, levels and angles
- PC14.** confirm that the coordinates of all the plotted points are as per the survey data
- PC15.** obtain approval from senior for confirming the correctness of the map
- PC16.** store and save the map as per organizational norms

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:



Qualification Pack

- KU1.** standard practices for surveying works
- KU2.** safety rules and regulations for handling and storing relevant tools, equipment, and materials required for relevant works in accordance with organizational norms
- KU3.** importance of personal protection including the use of related safety gears and equipment in accordance with organizational norms
- KU4.** service request procedures for tools, materials and equipment as per organizational norms
- KU5.** procedure for maintenance of tools and equipment
- KU6.** statutory compliance requirement related to workmen engagement
- KU7.** introduction to topographic survey and its types
- KU8.** different methods of conducting topographic survey
- KU9.** different instruments and their operation in conducting topographic surveying
- KU10.** how to plot the collected data to represent topography of the area in required scale
- KU11.** various software used in surveying and plotting
- KU12.** temporary and permanent adjustments of various equipment
- KU13.** application of different types of topographic survey in various sectors
- KU14.** different methods of computing levels, angles, bearing and distances using modern instruments
- KU15.** different methods of computing levels, angles, bearing and distances using conventional instruments
- KU16.** calculation and properties of contour lines and plotting the same with required scale
- KU17.** interpretation of contour lines
- KU18.** importance of contour lines

Generic Skills (GS)

User/individual on the job needs to know how to:

- GS1.** write in at least one language, preferably in the local language of the site
- GS2.** read sketches, instructions provided for the work, and various signboards, safety rules, safety tags, exit route information in one or more languages, preferably in the local language of the site
- GS3.** speak in one or more language, preferably one of the local language at the site
- GS4.** communicate orally and effectively with team members
- GS5.** analyze the safety aspect of the workplace
- GS6.** plan work and organize required resource effectively
- GS7.** complete work as per agreed time schedule and quality parameters
- GS8.** resolve any conflict within the teammates
- GS9.** evaluate the complexity of the tasks
- GS10.** identify any violation of safety norms during the work



Qualification Pack

Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Carry out topographic survey</i>	15	35	-	-
PC1. identify the boundaries of the area to be surveyed	-	-	-	-
PC2. establish control points/bench marks to initiate the survey	-	-	-	-
PC3. identify a suitable location for setting up a total station such that maximum points or features are visible from this point	-	-	-	-
PC4. set up the instrument at identified point and perform necessary adjustments	-	-	-	-
PC5. take the back sight from instrument point to BM and set the readings as zero for this point	-	-	-	-
PC6. instruct the subordinate to place the staff at all relevant feature points like corners of buildings, trees, pillars etc	-	-	-	-
PC7. operate the survey instrument to focus on the staff and record the required readings for the fore sights/intermediate sight as per the requirement	-	-	-	-
PC8. coordinate with team members to change the instrument placing and repeat the same procedure for taking further reading	-	-	-	-
PC9. transfer/upload the recorded data on the computer system using appropriate CAD software	-	-	-	-
<i>Prepare contour and topographic map</i>	15	35	-	-
PC10. obtain approval from seniors regarding correctness of the collected data	-	-	-	-
PC11. segregate the required information in desired formats using available software	-	-	-	-
PC12. decide the scale for plotting the survey data	-	-	-	-



Qualification Pack

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC13. plot the recorded points using the CAD software to get the required contour/ topographic map as per the measured distance, location, levels and angles	-	-	-	-
PC14. confirm that the coordinates of all the plotted points are as per the survey data	-	-	-	-
PC15. obtain approval from senior for confirming the correctness of the map	-	-	-	-
PC16. store and save the map as per organizational norms	-	-	-	-
NOS Total	30	70	-	-



Qualification Pack

National Occupational Standards (NOS) Parameters

NOS Code	CON/N0907
NOS Name	Carry out topographic survey
Sector	Construction
Sub-Sector	Real Estate and Infrastructure construction
Occupation	Surveying
NSQF Level	5.5
Credits	3
Version	4.0
Last Reviewed Date	08/05/2025
Next Review Date	30/04/2028
NSQC Clearance Date	08/05/2025



Qualification Pack

CON/N9002: Manage workplace for safe and healthy work environment

Description

This unit describes the skill and knowledge required to maintain a healthy & safe working environment for the group of people working under an individual.

Scope

The scope covers the following :

- Ensure effective implementation of health, safety and environment policies and procedures
- Identify and respond to risks / fire and emergencies associated with the work practices and workplace
- Ensure sanitization and infection control guidelines are followed at construction site

Elements and Performance Criteria

Ensure effective implementation of health, safety and environment policies and procedures

To be competent, the user/individual on the job must be able to:

- PC1.** implement safe handling and stacking methods at workplace /store
- PC2.** ensure the adequate availability and placing of safety and protection installations at site
- PC3.** ensure that safe access ways are available at work place for movement of workers and materials
- PC4.** ensure the safe use of tools and tackles by teammates as per work requirements
- PC5.** ensure ergonomic principles are adopted by the teammates while lifting and shifting of construction materials, tools and equipment
- PC6.** ensure appropriate use of following Personal Protective Equipment (PPE) as per work requirement of head protection, ear protection, fall protection, foot protection, face and eye protection, hand and body protection, respiratory protection
- PC7.** maintain entry and exit pathways from confined spaces, excavated pits and other location as per safety parameters/instructions
- PC8.** ensure proper housekeeping at workplace
- PC9.** ensure that subordinates adhere to health and safety plans

Identify and respond to risks / fire and emergencies associated with the work practices and workplace

To be competent, the user/individual on the job must be able to:

- PC10.** identify any hazard at workplace and report/notify the same to appropriate authorities.
- PC11.** follow procedures for accident recording and reporting as per organizational and statutory requirements
- PC12.** ensure effective adherence to emergency response procedures / protocols
- PC13.** select and operate different types of fire extinguishers corresponding to types of fires as per EHS guideline
- PC14.** obtain 'height pass' clearance as per EHS guideline



Qualification Pack

PC15. implement control measures to reduce risks ,meeting legislative requirements within the scope of own role and expertise, as per organizational policies

Ensure sanitization and infection control guidelines are followed at construction site

To be competent, the user/individual on the job must be able to:

PC16. promote awareness about latest hygiene and sanitation regulations

PC17. ensure disinfection procedure related to material, tools and supplies are followed properly

PC18. respond to infection prevention and control and its non-compliance , within scope of own role or report to required personnel

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

- KU1.** the policies, procedures and protocol set up by the EHS Department with respect to Health , Safety and Environment at the respective construction site
- KU2.** reporting procedures in cases of breaches or hazards in site safety, accidents or emergency situations
- KU3.** safe working practices for tools, tackles and equipment
- KU4.** workplace policies and health and safety requirements for dealing with potential risks as defined by the EHS department
- KU5.** how to respond to accidents and emergencies
- KU6.** the appropriate personal protective equipment to be used based on various working conditions
- KU7.** how to use necessary material ,tools, tackles and equipment in a safe and appropriate manner as specified by site EHS for each level and respective workman gang
- KU8.** ways of transmission of infection
- KU9.** ways to manage infectious risks at the workplace
- KU10.** different methods of cleaning, disinfection, sterilization and sanitization
- KU11.** symptoms of infection like fever, cough, redness, swelling and inflammation
- KU12.** actions be taken during a medical emergency
- KU13.** current guidelines, national legislation, local policies and protocols regarding spread of infectious disease.

Generic Skills (GS)

User/individual on the job needs to know how to:

- GS1.** write in at least one language, preferably in the local language of the site
- GS2.** read instructions provided for the work, and various signboards, safety rules, safety tags, exit route information in one or more languages, preferably in the local language of the site
- GS3.** speak in one or more language, preferably one of the local language at the site
- GS4.** listen and follow instructions shared by site EHS and superiors regarding site safety
- GS5.** communicate reporting of site conditions, hazards, accidents, etc.
- GS6.** analyze the safety aspect of the workplace



Qualification Pack

GS7. identify any violation of safety norms during the work



Qualification Pack

Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Ensure effective implementation of health, safety and environment policies and procedures</i>	15	40	-	-
PC1. implement safe handling and stacking methods at workplace /store	-	-	-	-
PC2. ensure the adequate availability and placing of safety and protection installations at site	-	-	-	-
PC3. ensure that safe access ways are available at work place for movement of workers and materials	-	-	-	-
PC4. ensure the safe use of tools and tackles by teammates as per work requirements	-	-	-	-
PC5. ensure ergonomic principles are adopted by the teammates while lifting and shifting of construction materials, tools and equipment	-	-	-	-
PC6. ensure appropriate use of following Personal Protective Equipment (PPE) as per work requirement of head protection, ear protection, fall protection, foot protection, face and eye protection, hand and body protection, respiratory protection	-	-	-	-
PC7. maintain entry and exit pathways from confined spaces, excavated pits and other location as per safety parameters/instructions	-	-	-	-
PC8. ensure proper housekeeping at workplace	-	-	-	-
PC9. ensure that subordinates adhere to health and safety plans	-	-	-	-
<i>Identify and respond to risks / fire and emergencies associated with the work practices and workplace</i>	10	20	-	-
PC10. identify any hazard at workplace and report/notify the same to appropriate authorities.	-	-	-	-
PC11. follow procedures for accident recording and reporting as per organizational and statutory requirements	-	-	-	-
PC12. ensure effective adherence to emergency response procedures / protocols	-	-	-	-



Qualification Pack

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC13. select and operate different types of fire extinguishers corresponding to types of fires as per EHS guideline	-	-	-	-
PC14. obtain 'height pass' clearance as per EHS guideline	-	-	-	-
PC15. implement control measures to reduce risks ,meeting legislative requirements within the scope of own role and expertise, as per organizational policies	-	-	-	-
<i>Ensure sanitization and infection control guidelines are followed at construction site</i>	5	10	-	-
PC16. promote awareness about latest hygiene and sanitation regulations	-	-	-	-
PC17. ensure disinfection procedure related to material, tools and supplies are followed properly	-	-	-	-
PC18. respond to infection prevention and control and its non-compliance , within scope of own role or report to required personnel	-	-	-	-
NOS Total	30	70	-	-



Qualification Pack

National Occupational Standards (NOS) Parameters

NOS Code	CON/N9002
NOS Name	Manage workplace for safe and healthy work environment
Sector	Construction
Sub-Sector	Generic
Occupation	Generic Safety
NSQF Level	5
Credits	1
Version	4.0
Last Reviewed Date	08/05/2025
Next Review Date	31/10/2025
NSQC Clearance Date	08/05/2025



Qualification Pack

CON/N8003: Supervise, monitor and evaluate performance of subordinates at workplace

Description

This unit covers the skills and knowledge required by an individual to supervise, monitor and evaluate performance of subordinates at the work place

Scope

The scope covers the following :

- Monitor and evaluate subordinates performance as per quality standards and timelines
- Practice and promote gender inclusive behavior and leadership

Elements and Performance Criteria

Monitor and evaluate subordinates performance as per quality standards and timelines

To be competent, the user/individual on the job must be able to:

- PC1.** fix expected targets for the respective gang as per site requirements and allocate work to subordinates
- PC2.** establish expected performance standards and expectations for the respective gang of workers to meet the desired outcomes
- PC3.** inspect assigned work to the respected gang of workers through progressive checking
- PC4.** observe and verify the work activities performed by the subordinates at the construction site
- PC5.** monitor overall performance of subordinates on the designed measures to ensure quality requirements set by the concerned authority
- PC6.** ensure adherence to the organizational policies and procedures for all relevant construction activities by the workmen subordinations

Practice and promote gender inclusive behavior and leadership

To be competent, the user/individual on the job must be able to:

- PC7.** promote a safe, supportive, inclusive and gender sensitive environment at workplace.
- PC8.** manage and facilitate capacity building to enable inclusion of all genders
- PC9.** take prompt corrective action in case of inappropriate behavior at workplace

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

- KU1.** policies, procedures and work targets for performance evaluation and appraisals
- KU2.** organizational policies, procedures and protocol for smooth completion of work at the respective workplace
- KU3.** complete work/task accurately by following standard specifications and procedures by optimized and correct used of materials , tools, tackles and equipment



Qualification Pack

Generic Skills (GS)

User/individual on the job needs to know how to:

- GS1.** write in at one or more language, preferably the local language
- GS2.** read in at one or more language, preferably the local language at the site
- GS3.** speak in at one or more language, preferably one of the local language at the site
- GS4.** communicate effectively with team members at workplace
- GS5.** plan work as per right sequence and organize required resources in coordination with the team members
- GS6.** complete work as per the requirement
- GS7.** evaluate and find solutions to minimize errors and suggest improvements for optimizing resource utilization
- GS8.** assess which situations would require intervention of superiors



Qualification Pack

Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Monitor and evaluate subordinates performance as per quality standards and timelines</i>	20	40	-	-
PC1. fix expected targets for the respective gang as per site requirements and allocate work to subordinates	-	-	-	-
PC2. establish expected performance standards and expectations for the respective gang of workers to meet the desired outcomes	-	-	-	-
PC3. inspect assigned work to the respected gang of workers through progressive checking	-	-	-	-
PC4. observe and verify the work activities performed by the subordinates at the construction site	-	-	-	-
PC5. monitor overall performance of subordinates on the designed measures to ensure quality requirements set by the concerned authority	-	-	-	-
PC6. ensure adherence to the organizational policies and procedures for all relevant construction activities by the workmen subordinations	-	-	-	-
<i>Practice and promote gender inclusive behavior and leadership</i>	10	30	-	-
PC7. promote a safe, supportive, inclusive and gender sensitive environment at workplace.	-	-	-	-
PC8. manage and facilitate capacity building to enable inclusion of all genders	-	-	-	-
PC9. take prompt corrective action in case of inappropriate behavior at workplace	-	-	-	-
NOS Total	30	70	-	-



Qualification Pack

National Occupational Standards (NOS) Parameters

NOS Code	CON/N8003
NOS Name	Supervise, monitor and evaluate performance of subordinates at workplace
Sector	Construction
Sub-Sector	Real Estate and Infrastructure construction
Occupation	Generic 2
NSQF Level	5.5
Credits	1
Version	4.0
Last Reviewed Date	08/05/2025
Next Review Date	31/10/2025
NSQC Clearance Date	08/05/2025



Qualification Pack

DGT/VSQ/N0103: Employability Skills (90 Hours)

Description

This unit is about employability skills, Constitutional values, becoming a professional in the 21st Century, digital, financial, and legal literacy, diversity and Inclusion, English and communication skills, customer service, entrepreneurship, and apprenticeship, getting ready for jobs and career development.

Scope

The scope covers the following :

- Introduction to Employability Skills
- Constitutional values - Citizenship
- Becoming a Professional in the 21st Century
- Basic English Skills
- Career Development & Goal Setting
- Communication Skills
- Diversity & Inclusion
- Financial and Legal Literacy
- Essential Digital Skills
- Entrepreneurship
- Customer Service
- Getting ready for Apprenticeship & Jobs

Elements and Performance Criteria

Introduction to Employability Skills

To be competent, the user/individual on the job must be able to:

- PC1.** understand the significance of employability skills in meeting the current job market requirement and future of work
- PC2.** identify and explore learning and employability relevant portals
- PC3.** research about the different industries, job market trends, latest skills required and the available opportunities

Constitutional values - Citizenship

To be competent, the user/individual on the job must be able to:

- PC4.** recognize the significance of constitutional values, including civic rights and duties, citizenship, responsibility towards society etc. and personal values and ethics such as honesty, integrity, caring and respecting others, etc.
- PC5.** follow environmentally sustainable practices

Becoming a Professional in the 21st Century

To be competent, the user/individual on the job must be able to:

- PC6.** recognize the significance of 21st Century Skills for employment



Qualification Pack

- PC7.** practice the 21st Century Skills such as Self-Awareness, Behaviour Skills, time management, critical and adaptive thinking, problem-solving, creative thinking, social and cultural awareness, emotional awareness, learning to learn for continuous learning etc. in personal and professional life
- PC8.** adopt a continuous learning mindset for personal and professional development

Basic English Skills

To be competent, the user/individual on the job must be able to:

- PC9.** use basic English for everyday conversation in different contexts, in person and over the telephone
- PC10.** read and understand routine information, notes, instructions, mails, letters etc. written in English
- PC11.** write short messages, notes, letters, e-mails etc. in English

Career Development & Goal Setting

To be competent, the user/individual on the job must be able to:

- PC12.** identify career goals based on the skills, interests, knowledge, and personal attributes
- PC13.** prepare a career development plan with short- and long-term goals

Communication Skills

To be competent, the user/individual on the job must be able to:

- PC14.** follow verbal and non-verbal communication etiquette while communicating in professional and public settings
- PC15.** use active listening techniques for effective communication
- PC16.** communicate in writing using appropriate style and format based on formal or informal requirements
- PC17.** work collaboratively with others in a team

Diversity & Inclusion

To be competent, the user/individual on the job must be able to:

- PC18.** communicate and behave appropriately with all genders and PwD
- PC19.** escalate any issues related to sexual harassment at workplace according to POSH Act

Financial and Legal Literacy

To be competent, the user/individual on the job must be able to:

- PC20.** identify and select reliable institutions for various financial products and services such as bank account, debit and credit cards, loans, insurance etc.
- PC21.** carry out offline and online financial transactions, safely and securely, using various methods and check the entries in the passbook
- PC22.** identify common components of salary and compute income, expenses, taxes, investments etc
- PC23.** identify relevant rights and laws and use legal aids to fight against legal exploitation

Essential Digital Skills

To be competent, the user/individual on the job must be able to:

- PC24.** operate digital devices and use their features and applications securely and safely
- PC25.** carry out basic internet operations by connecting to the internet safely and securely, using the mobile data or other available networks through Bluetooth, Wi-Fi, etc.
- PC26.** display responsible online behaviour while using various social media platforms



Qualification Pack

- PC27.** create a personal email account, send and process received messages as per requirement
- PC28.** carry out basic procedures in documents, spreadsheets and presentations using respective and appropriate applications
- PC29.** utilize virtual collaboration tools to work effectively

Entrepreneurship

To be competent, the user/individual on the job must be able to:

- PC30.** identify different types of Entrepreneurship and Enterprises and assess opportunities for potential business through research
- PC31.** develop a business plan and a work model, considering the 4Ps of Marketing Product, Price, Place and Promotion
- PC32.** identify sources of funding, anticipate, and mitigate any financial/ legal hurdles for the potential business opportunity

Customer Service

To be competent, the user/individual on the job must be able to:

- PC33.** identify different types of customers and ways to communicate with them
- PC34.** identify and respond to customer requests and needs in a professional manner
- PC35.** use appropriate tools to collect customer feedback
- PC36.** follow appropriate hygiene and grooming standards

Getting ready for apprenticeship & Jobs

To be competent, the user/individual on the job must be able to:

- PC37.** create a professional Curriculum vitae (Résumé)
- PC38.** search for suitable jobs using reliable offline and online sources such as Employment exchange, recruitment agencies, newspapers etc. and job portals, respectively
- PC39.** apply to identified job openings using offline /online methods as per requirement
- PC40.** answer questions politely, with clarity and confidence, during recruitment and selection
- PC41.** identify apprenticeship opportunities and register for it as per guidelines and requirements

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

- KU1.** need for employability skills and different learning and employability related portals
- KU2.** various constitutional and personal values
- KU3.** different environmentally sustainable practices and their importance
- KU4.** Twenty first (21st) century skills and their importance
- KU5.** how to use English language for effective verbal (face to face and telephonic) and written communication in formal and informal set up
- KU6.** importance of career development and setting long- and short-term goals
- KU7.** about effective communication
- KU8.** POSH Act
- KU9.** Gender sensitivity and inclusivity
- KU10.** different types of financial institutes, products, and services



Qualification Pack

- KU11.** components of salary and how to compute income and expenditure
- KU12.** importance of maintaining safety and security in offline and online financial transactions
- KU13.** different legal rights and laws
- KU14.** different types of digital devices and the procedure to operate them safely and securely
- KU15.** how to create and operate an e- mail account
- KU16.** use applications such as word processors, spreadsheets etc.
- KU17.** how to identify business opportunities
- KU18.** types and needs of customers
- KU19.** how to apply for a job and prepare for an interview
- KU20.** apprenticeship scheme and the process of registering on apprenticeship portal

Generic Skills (GS)

User/individual on the job needs to know how to:

- GS1.** read and write different types of documents/instructions/correspondence in English and other languages
- GS2.** communicate effectively using appropriate language in formal and informal settings
- GS3.** behave politely and appropriately with all to maintain effective work relationship
- GS4.** how to work in a virtual mode, using various technological platforms
- GS5.** perform calculations efficiently
- GS6.** solve problems effectively
- GS7.** pay attention to details
- GS8.** manage time efficiently
- GS9.** maintain hygiene and sanitization to avoid infection



Qualification Pack

Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Introduction to Employability Skills</i>	1	1	-	-
PC1. understand the significance of employability skills in meeting the current job market requirement and future of work	-	-	-	-
PC2. identify and explore learning and employability relevant portals	-	-	-	-
PC3. research about the different industries, job market trends, latest skills required and the available opportunities	-	-	-	-
<i>Constitutional values – Citizenship</i>	1	1	-	-
PC4. recognize the significance of constitutional values, including civic rights and duties, citizenship, responsibility towards society etc. and personal values and ethics such as honesty, integrity, caring and respecting others, etc.	-	-	-	-
PC5. follow environmentally sustainable practices	-	-	-	-
<i>Becoming a Professional in the 21st Century</i>	1	3	-	-
PC6. recognize the significance of 21st Century Skills for employment	-	-	-	-
PC7. practice the 21st Century Skills such as Self-Awareness, Behaviour Skills, time management, critical and adaptive thinking, problem-solving, creative thinking, social and cultural awareness, emotional awareness, learning to learn for continuous learning etc. in personal and professional life	-	-	-	-
PC8. adopt a continuous learning mindset for personal and professional development	-	-	-	-
<i>Basic English Skills</i>	3	4	-	-
PC9. use basic English for everyday conversation in different contexts, in person and over the telephone	-	-	-	-



Qualification Pack

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC10. read and understand routine information, notes, instructions, mails, letters etc. written in English	-	-	-	-
PC11. write short messages, notes, letters, e-mails etc. in English	-	-	-	-
<i>Career Development & Goal Setting</i>	1	2	-	-
PC12. identify career goals based on the skills, interests, knowledge, and personal attributes	-	-	-	-
PC13. prepare a career development plan with short- and long-term goals	-	-	-	-
<i>Communication Skills</i>	2	2	-	-
PC14. follow verbal and non-verbal communication etiquette while communicating in professional and public settings	-	-	-	-
PC15. use active listening techniques for effective communication	-	-	-	-
PC16. communicate in writing using appropriate style and format based on formal or informal requirements	-	-	-	-
PC17. work collaboratively with others in a team	-	-	-	-
<i>Diversity & Inclusion</i>	1	1	-	-
PC18. communicate and behave appropriately with all genders and PwD	-	-	-	-
PC19. escalate any issues related to sexual harassment at workplace according to POSH Act	-	-	-	-
<i>Financial and Legal Literacy</i>	2	3	-	-
PC20. identify and select reliable institutions for various financial products and services such as bank account, debit and credit cards, loans, insurance etc.	-	-	-	-
PC21. carry out offline and online financial transactions, safely and securely, using various methods and check the entries in the passbook	-	-	-	-



Qualification Pack

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC22. identify common components of salary and compute income, expenses, taxes, investments etc	-	-	-	-
PC23. identify relevant rights and laws and use legal aids to fight against legal exploitation	-	-	-	-
<i>Essential Digital Skills</i>	3	5	-	-
PC24. operate digital devices and use their features and applications securely and safely	-	-	-	-
PC25. carry out basic internet operations by connecting to the internet safely and securely, using the mobile data or other available networks through Bluetooth, Wi-Fi, etc.	-	-	-	-
PC26. display responsible online behaviour while using various social media platforms	-	-	-	-
PC27. create a personal email account, send and process received messages as per requirement	-	-	-	-
PC28. carry out basic procedures in documents, spreadsheets and presentations using respective and appropriate applications	-	-	-	-
PC29. utilize virtual collaboration tools to work effectively	-	-	-	-
<i>Entrepreneurship</i>	2	3	-	-
PC30. identify different types of Entrepreneurship and Enterprises and assess opportunities for potential business through research	-	-	-	-
PC31. develop a business plan and a work model, considering the 4Ps of Marketing Product, Price, Place and Promotion	-	-	-	-
PC32. identify sources of funding, anticipate, and mitigate any financial/ legal hurdles for the potential business opportunity	-	-	-	-
<i>Customer Service</i>	1	2	-	-
PC33. identify different types of customers and ways to communicate with them	-	-	-	-



Qualification Pack

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC34. identify and respond to customer requests and needs in a professional manner	-	-	-	-
PC35. use appropriate tools to collect customer feedback	-	-	-	-
PC36. follow appropriate hygiene and grooming standards	-	-	-	-
<i>Getting ready for apprenticeship & Jobs</i>	2	3	-	-
PC37. create a professional Curriculum vitae (Résumé)	-	-	-	-
PC38. search for suitable jobs using reliable offline and online sources such as Employment exchange, recruitment agencies, newspapers etc. and job portals, respectively	-	-	-	-
PC39. apply to identified job openings using offline /online methods as per requirement	-	-	-	-
PC40. answer questions politely, with clarity and confidence, during recruitment and selection	-	-	-	-
PC41. identify apprenticeship opportunities and register for it as per guidelines and requirements	-	-	-	-
NOS Total	20	30	-	-



Qualification Pack

National Occupational Standards (NOS) Parameters

NOS Code	DGT/VSQ/N0103
NOS Name	Employability Skills (90 Hours)
Sector	Cross Sectoral
Sub-Sector	Professional Skills
Occupation	Employability
NSQF Level	5
Credits	3
Version	1.0
Last Reviewed Date	07/10/2025
Next Review Date	07/10/2028
NSQC Clearance Date	07/10/2025

Assessment Guidelines and Assessment Weightage

Assessment Guidelines

1. Criteria for assessment for each Qualification Pack will be created by CSDCI. Each Performance Criteria (PC)/ Elements will be assigned marks proportional to its importance in NOS.
2. CSDCI will also lay down proportion of marks for Theory and Skills Practical for each PC/ Elements. The assessment for the knowledge part will be based on knowledge bank of questions created by Assessment Bodies subject to approval by CSDCI.
3. Individual assessment agencies will create unique question papers for knowledge/theory part for assessment of candidates as per assessment criteria given below:
 - a) Individual assessment agencies will create unique evaluations for skill practical for every student at each examination/training center based on assessment criteria.
 - b) The passing percentage for the QP will be 70%. To pass the Qualification Pack.
 - c) The Assessor shall check the final outcome of the practices while evaluating the steps performed to achieve the final outcome
 - d) The trainee shall be provided with a chance to repeat the test to correct his procedures in case of



Qualification Pack

improper performance, with a deduction of marks for each iteration.

e) After the certain number of iteration as decided by SSC the trainee is marked as fail, scoring zero marks for the procedure for the practical activity.

f) In case of successfully passing only certain number of NOS's, the trainee is eligible to take subsequent assessment on the balance NOS's to pass the Qualification Pack within the specified time frame set by CSDCI.

g) The candidates shall undergo On job training (OJT), and will learn at actual workplace for a fixed period of time and a certain weightage of assessment is allocated out of total skill weightage of Qualification Pack for undergoing OJT as stipulated by CSDCI. This OJT score and assessors' end point score are combined to arrive at final Marking/grading of trainees' skill test. The OJT score is determined by Supervisor of company under which candidates undergo on job training.

Minimum Aggregate Passing % at QP Level : 70

(Please note: Every Trainee should score a minimum aggregate passing percentage as specified above, to successfully clear the Qualification Pack assessment.)

Assessment Weightage

Compulsory NOS

National Occupational Standards	Theory Marks	Practical Marks	Project Marks	Viva Marks	Total Marks	Weightage
CON/N0903.Carry out temporary adjustment of survey instruments as per standard methods	30	70	-	-	100	15
CON/N0904.Conduct linear measurements using survey instruments and tools	30	70	-	-	100	15
CON/N0905.Carry out leveling and cross sectioning survey	30	70	-	-	100	15
CON/N0906.Carry out setting out operations for buildings/ other structures	30	70	-	-	100	15
CON/N0907.Carry out topographic survey	30	70	-	-	100	15



Qualification Pack

National Occupational Standards	Theory Marks	Practical Marks	Project Marks	Viva Marks	Total Marks	Weightage
CON/N9002.Manage workplace for safe and healthy work environment	30	70	-	-	100	10
CON/N8003.Supervise, monitor and evaluate performance of subordinates at workplace	30	70	-	-	100	10
DGT/VSQ/N0103.Employability Skills (90 Hours)	20	30	-	-	50	5
Total	230	520	-	-	750	100



Qualification Pack

Acronyms

NOS	National Occupational Standard(s)
NSQF	National Skills Qualifications Framework
QP	Qualifications Pack
TVET	Technical and Vocational Education and Training
MSDE	Ministry of Skill Development and Entrepreneurship
NCVET	National Council for Vocational Education and Training
NSDC	National Skill Development Corporation
SIDH	Skill India Digital Hub
CSDCI	Construction Skill Development Council of India
AB	Awarding Body
SSC	Sector Skill Council
PMKVY	Pradhan Mantri Kaushal Vikas Yojana
DDU-GKY	Deen Dayal Upadhyaya Grameen Kaushalya Yojana
SANKALP	Skill Acquisition and Knowledge Awareness for Livelihood Promotion
STRIVE	Skills Strengthening for Industrial Value Enhancement
JSS	Jan Shikshan Sansthan
STT	Short Term Training
RPL	Recognition of Prior Learning
NAPS	National Apprenticeship Promotion Scheme
AA	Assessment Agency
TP	Training Provider / Training Partner
TC	Training Centre
ITI	Industrial Training Institute
NSQC	National Skill Qualification Committee
NSQF	National Skills Qualification Framework



Qualification Pack

Q-File	Qualification File
QP	Qualification Pack
MC	Model Curriculum
NOS	National Occupational Standards
PC	Performance Criteria
KU	Knowledge and Understanding
GS	Generic Skills
MCQ	Multiple Choice Question
EHS	Environment Health and Safety
PPE	Personal Protective Equipment
QA/QC	Quality Assurance / Quality Control
TS	Total Station
GPS	Global Positioning System
DGPS	Differential Global Positioning System
GNSS	Global Navigation Satellite System
GIS	Geographic Information System
RS	Remote Sensing
UAV	Unmanned Aerial Vehicle (Drone Surveying)
LDR	Light Detection and Ranging
DEM	Digital Elevation Model
DTM	Digital Terrain Model
DSM	Digital Surface Model
GCP	Ground Control Point
HFL	High Flood Level (Hydrographic Survey)
TBM	Temporary Bench Mar
BM	Bench Mark



Qualification Pack

FS	Foresight (in leveling)
BS	Backsight (in leveling)
IS	Intermediate Sight (in leveling)
NTS	National Topographic Series (Survey Maps)
RL	Reduced Level
LOS	Line of Sight
CL	Center Line (Road Survey)
ROW	Right of Way (Land Acquisition)
KML	Keyhole Markup Language (GIS & Mapping)
CAD	Computer-Aided Design
BIM	Building Information Modeling
DLR	Digital Land Record
ROR	Record of Right
FMB	Field Measurement Book
RSR	Revenue Survey Record
LPM	Land Parcel Mapping
GTS	Great Trigonometrical Survey
MSL	Mean Sea Level
NWL	Normal Water Level
SPM	Single Point Mooring (Hydrographic Survey)
EIA	Environmental Impact Assessment



Qualification Pack

Glossary

Sector	Sector is a conglomeration of different business operations having similar business and interests. It may also be defined as a distinct subset of the economy whose components share similar characteristics and interests.
Sub-sector	Sub-sector is derived from a further breakdown based on the characteristics and interests of its components.
Occupation	Occupation is a set of job roles, which perform similar/ related set of functions in an industry.
Job role	Job role defines a unique set of functions that together form a unique employment opportunity in an organisation.
Occupational Standards (OS)	OS specify the standards of performance an individual must achieve when carrying out a function in the workplace, together with the Knowledge and Understanding (KU) they need to meet that standard consistently. Occupational Standards are applicable both in the Indian and global contexts.
Performance Criteria (PC)	Performance Criteria (PC) are statements that together specify the standard of performance required when carrying out a task.
National Occupational Standards (NOS)	NOS are occupational standards which apply uniquely in the Indian context.
Qualifications Pack (QP)	QP comprises the set of OS, together with the educational, training and other criteria required to perform a job role. A QP is assigned a unique qualifications pack code.
Unit Code	Unit code is a unique identifier for an Occupational Standard, which is denoted by an 'N'
Unit Title	Unit title gives a clear overall statement about what the incumbent should be able to do.
Description	Description gives a short summary of the unit content. This would be helpful to anyone searching on a database to verify that this is the appropriate OS they are looking for.
Scope	Scope is a set of statements specifying the range of variables that an individual may have to deal with in carrying out the function which have a critical impact on quality of performance required.
Knowledge and Understanding (KU)	Knowledge and Understanding (KU) are statements which together specify the technical, generic, professional and organisational specific knowledge that an individual needs in order to perform to the required standard.



Qualification Pack

Organisational Context	Organisational context includes the way the organisation is structured and how it operates, including the extent of operative knowledge managers have of their relevant areas of responsibility.
Technical Knowledge	Technical knowledge is the specific knowledge needed to accomplish specific designated responsibilities.
Core Skills/ Generic Skills (GS)	Core skills or Generic Skills (GS) are a group of skills that are the key to learning and working in today's world. These skills are typically needed in any work environment in today's world. These skills are typically needed in any work environment. In the context of the OS, these include communication related skills that are applicable to most job roles.
Electives	Electives are NOS/set of NOS that are identified by the sector as contributive to specialization in a job role. There may be multiple electives within a QP for each specialized job role. Trainees must select at least one elective for the successful completion of a QP with Electives.
Options	Options are NOS/set of NOS that are identified by the sector as additional skills. There may be multiple options within a QP. It is not mandatory to select any of the options to complete a QP with Options.
National Occupational Standard	NOS defines the measurable performance outcomes required from an individual engaged in a particular task. They list down what an individual performing that task should know and also do.
Qualification	A formal outcome of an assessment and validation process which is obtained when a competent body determines that an individual has achieved learning outcomes to given standards
Qualification File	A Qualification File is a template designed to capture necessary information of a Qualification from the perspective of NSQF compliance. The Qualification File will be normally submitted by the awarding body for the qualification.
Sector	A grouping of professional activities on the basis of their main economic function, product, service or technology.
Long Term Training	Long-term skilling means any vocational training program undertaken for a year and above. https://ncvet.gov.in/sites/default/files/NCVET.pdf